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College of Science and Technology
Rinchending: Bhutan

DBS101
Database Systems Fundamentals
(SS2026)

Practical 1 Report

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Date: 13th March 2026

RUB Wheel of Academic Law: Academic Dishonesty

Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorised group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication**: the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration**: making a false declaration to receive special consideration by an Examination Board or to obtain extensions to deadlines or exemption from work.
5. **Falsification of data**: presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism**: the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

Aim

The aim of this practical exercise is to install the MYSQL Database Management System on a Microsoft Windows computer and understand the basic process of interacting with the database using SQL commands. The exercise provides students with hands-on experience in setting up a database environment, configuring the MySQL server, and verifying the installation by executing simple SQL commands through the command line interface(cli).

Theory

A Database Management System (DBMS) is software used to store, organize, retrieve and manage data efficiently. It allows users to create databases, manage large amounts of information and ensure that data is secure, consistent and accessible. DBMS helps reduce data redundancy, improves data integrity and accessible to multiple users with the same data simultaneously. (Introduction of DBMS, 2026)

One of the most widely used language for interacting with database is Structured Query Language (SQL). It is used to create database structures, insert new data, retrieve existing data, update records, and delete unwanted information.

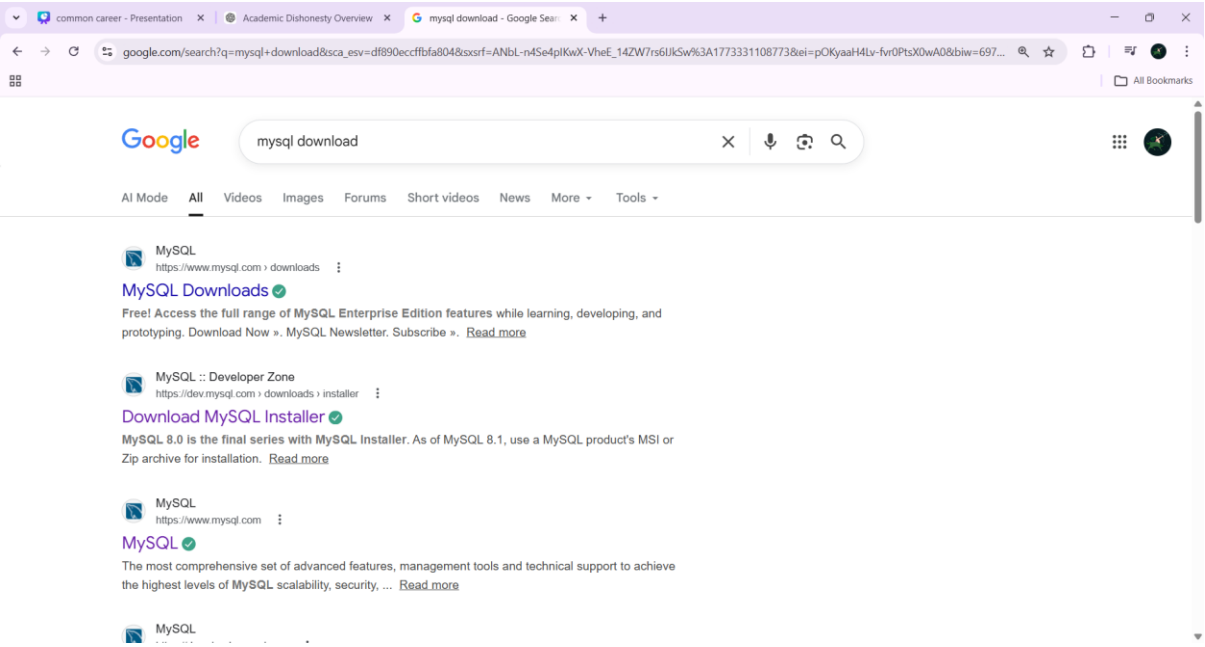
Another well-known database system is Oracle Database, which is widely used in large organizations because it provides advanced features such as strong security, high performance, cloud integration and support for large-scale enterprise data management. (Introduction to Oracle SQL)

Similarly, SAP is an enterprise software system used by businesses to manage important organizational processes such as finance, human resources, supply chain management, and customer relations. SAP integrates multiple business functions into a single platform allowing organizations to analyze data. (Definition and Meaning)

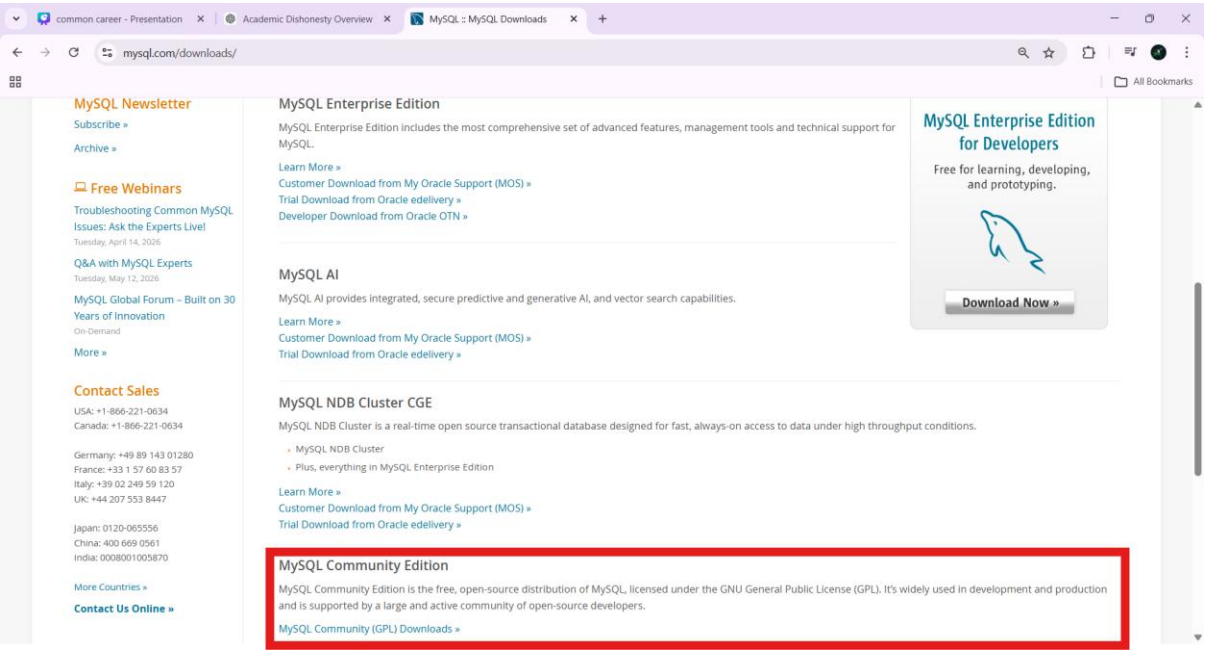
Together, systems such as MySQL, Oracle, and SAP demonstrate how database technologies are used in different environment that is from small software application and websites to large enterprise-level systems that manage complex business operations.

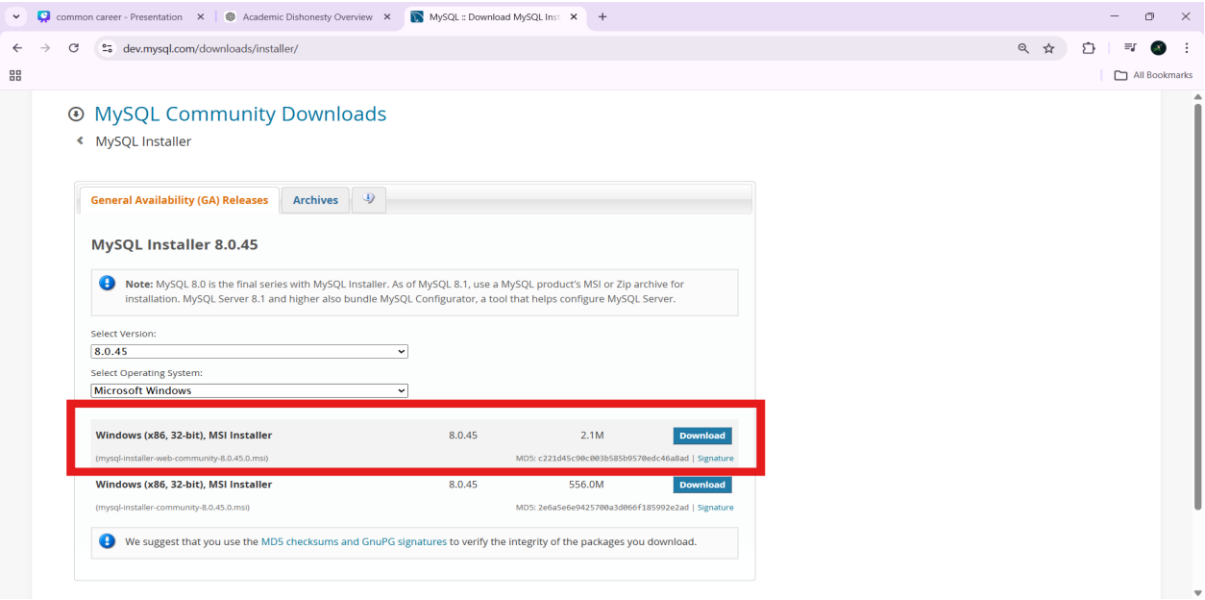
Implementation Steps

Task 1: Open web browser and search for MySQL download.

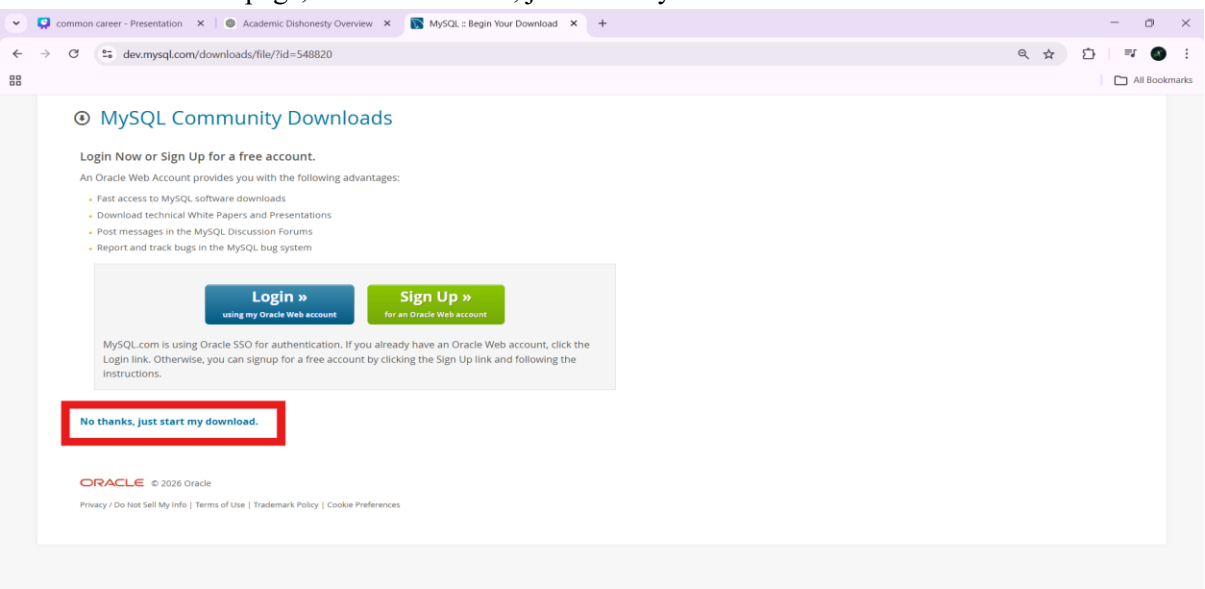


Task 2: Go to the Community Edition download page and select the 2.1 MB MySQL Installer.

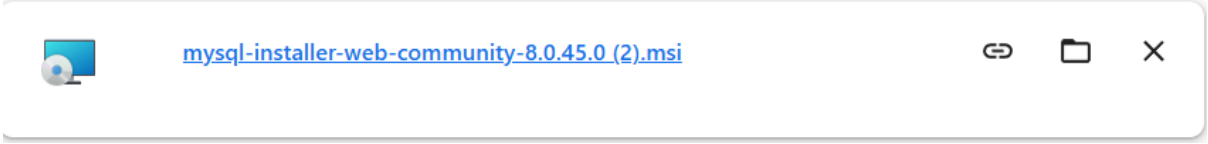




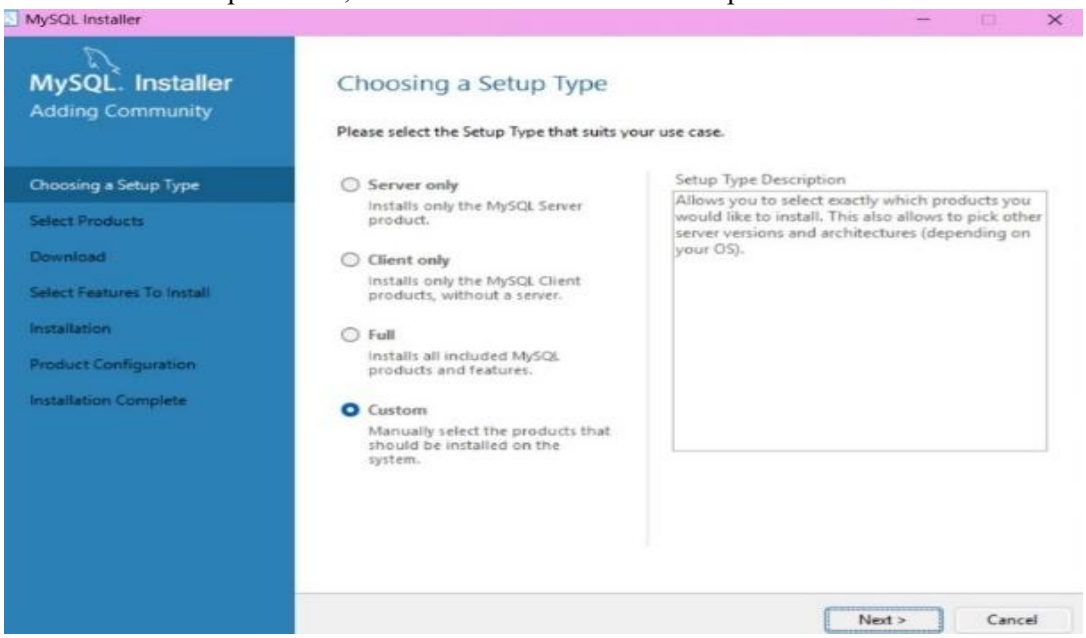
Task 3: On the next page, click "No thanks, just start my download."



Task 5: Open the downloaded folder and run the installer file.

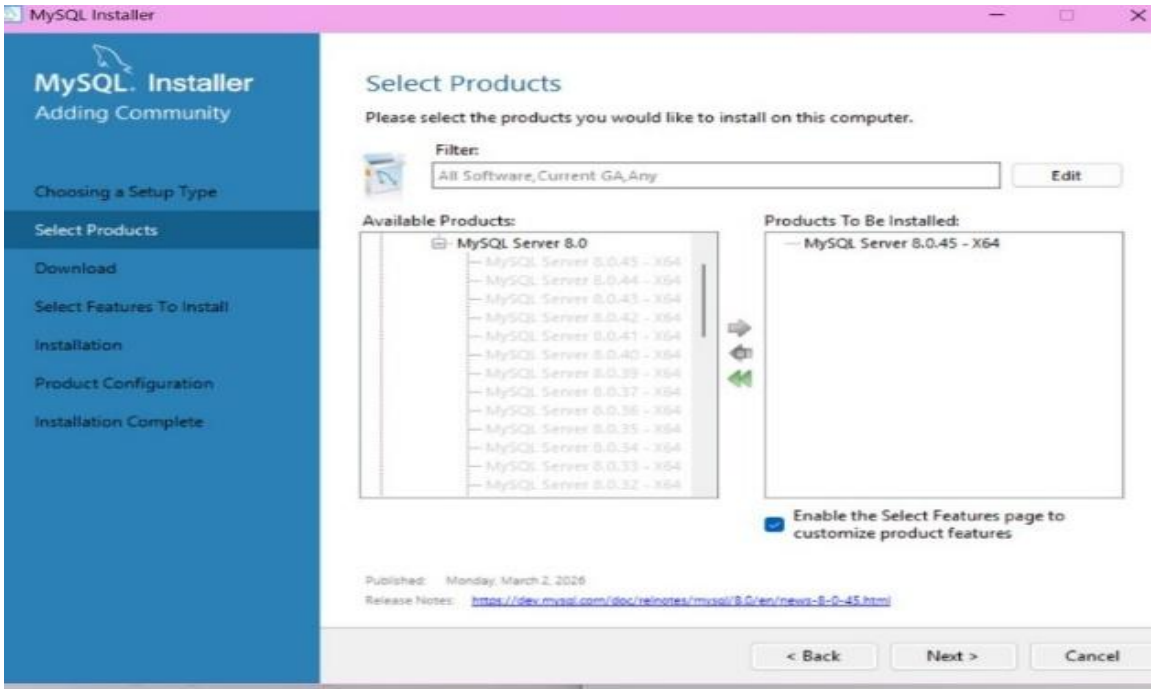


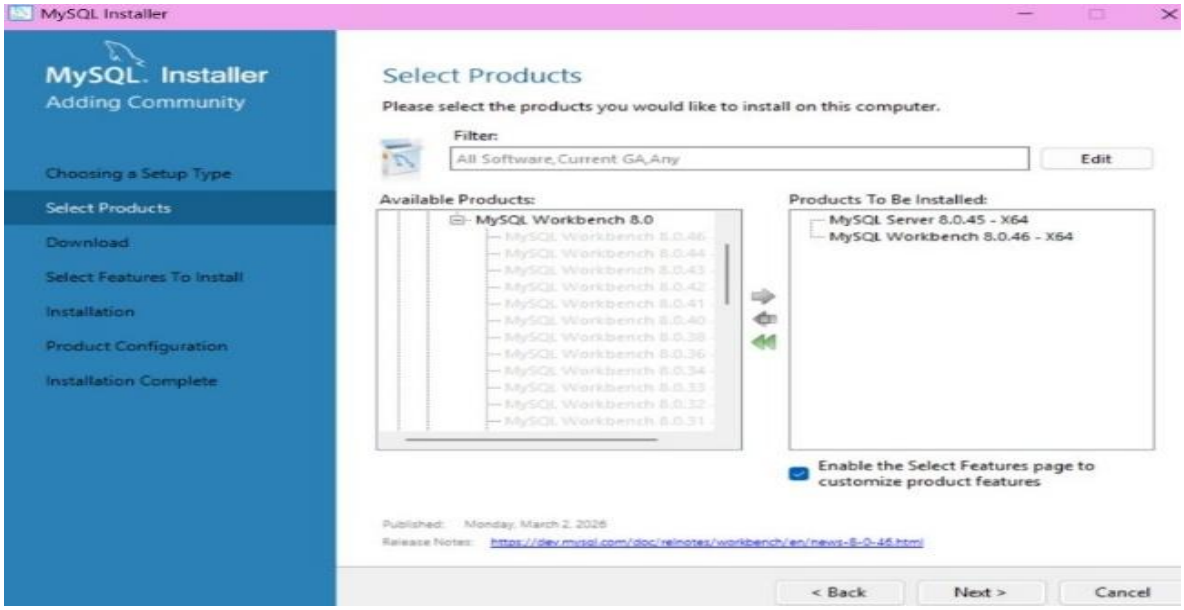
Task 6: In the setup window, select the Custom installation option and click Next.

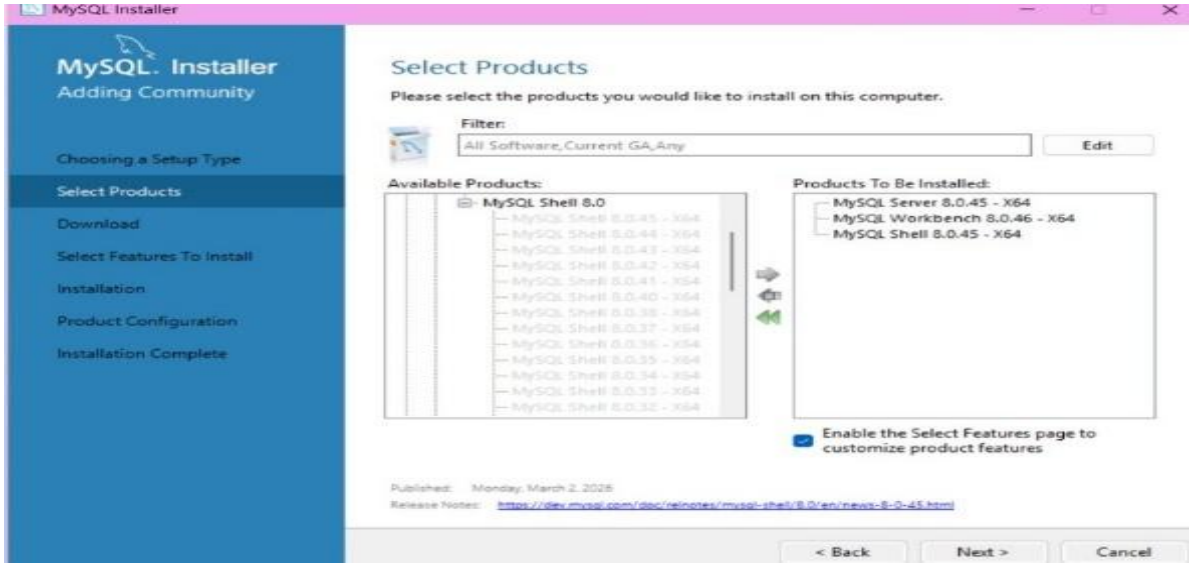


Task 7:

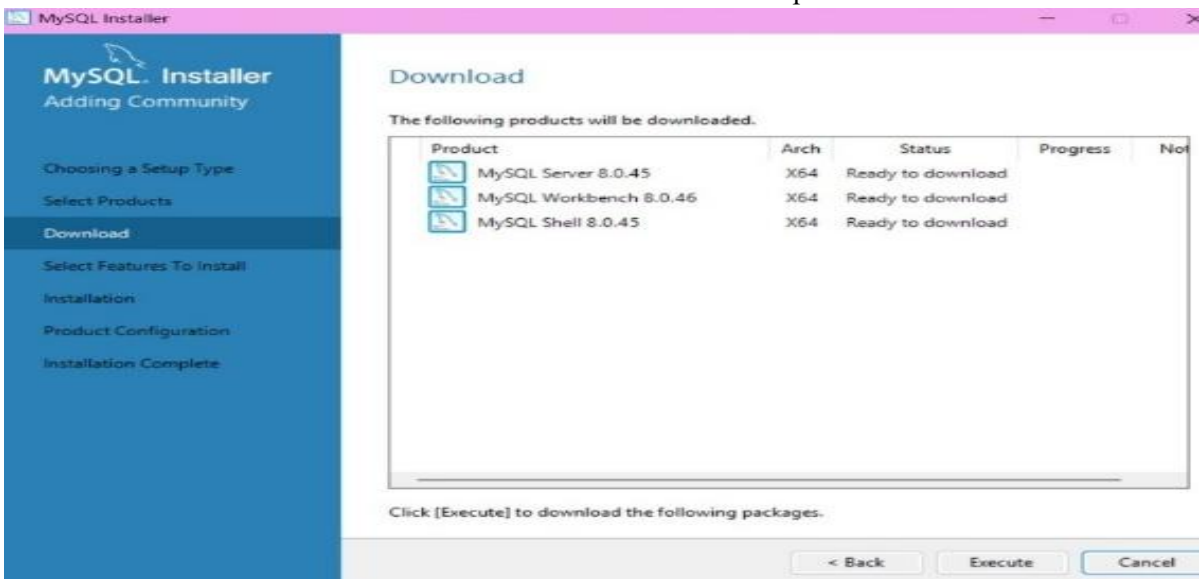
- Under MySQL Servers, select MySQL version 8.0.45 and move it to the right side.
- Under Applications, select MySQL Workbench version 8.0.46 and move it to the right side.
- Again, under Applications, select MySQL Shell version 8.0.45 and move it to the right side.



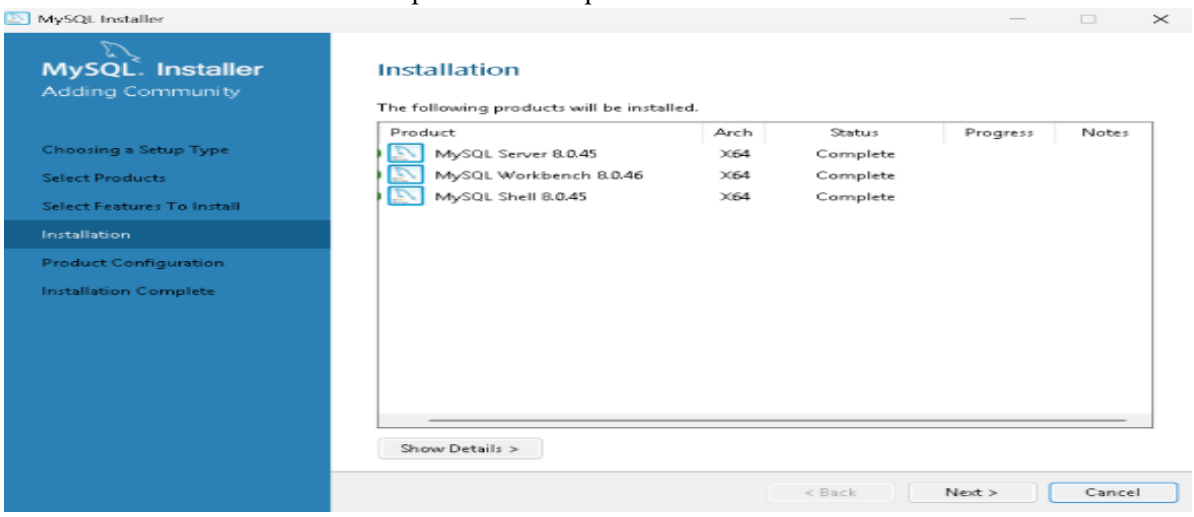




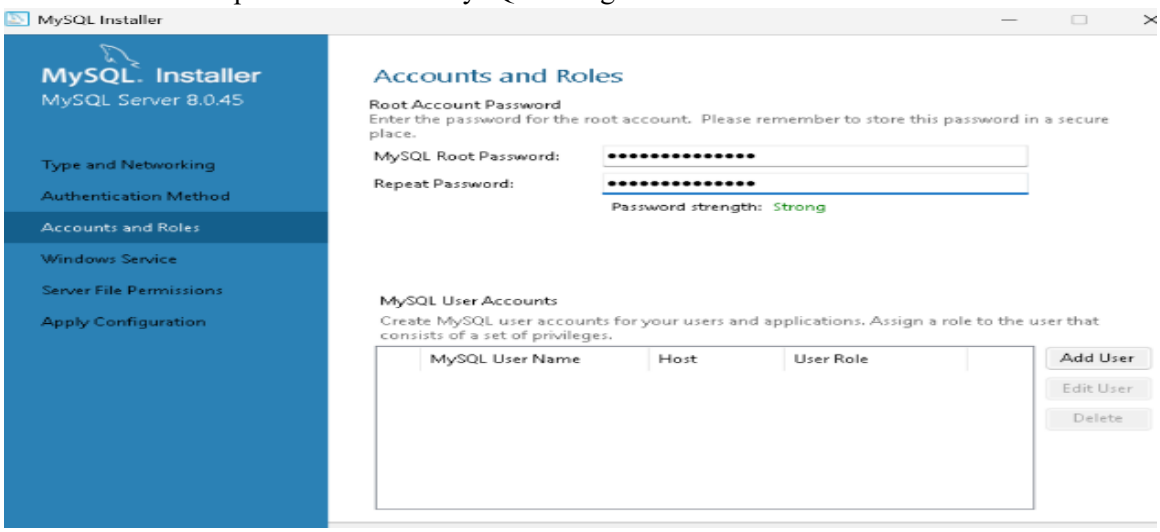
Task 8: Proceed with the installation with these three selected components.



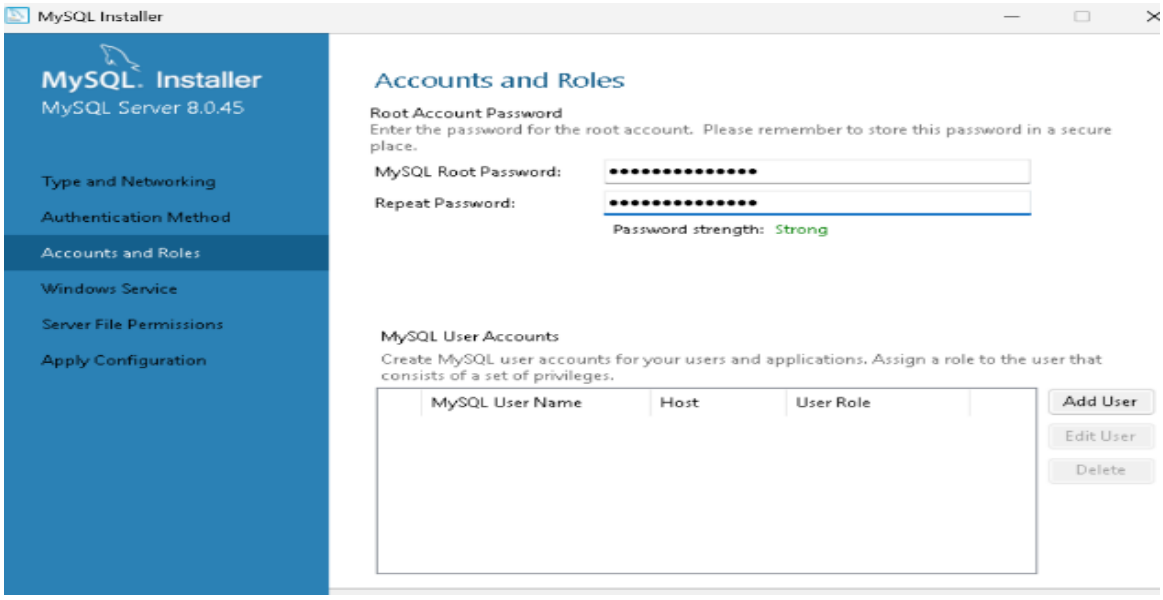
Task 9: Wait until the installation process is completed and click Next.



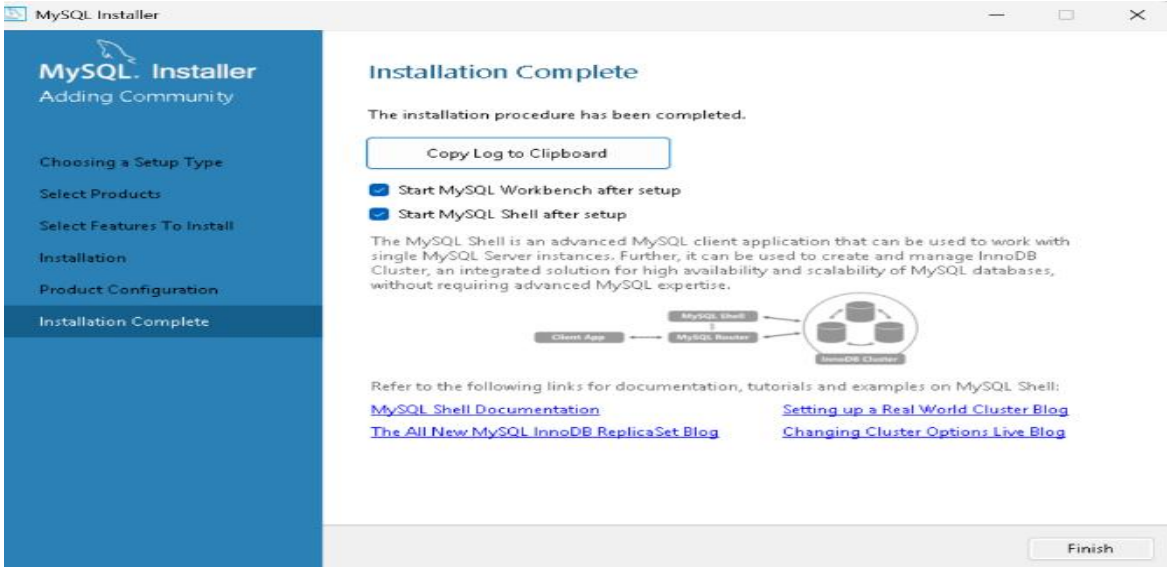
Task 10: Enter the password for the MySQL configuration and click Next.



Task 11: Click Execute to apply the configuration settings.

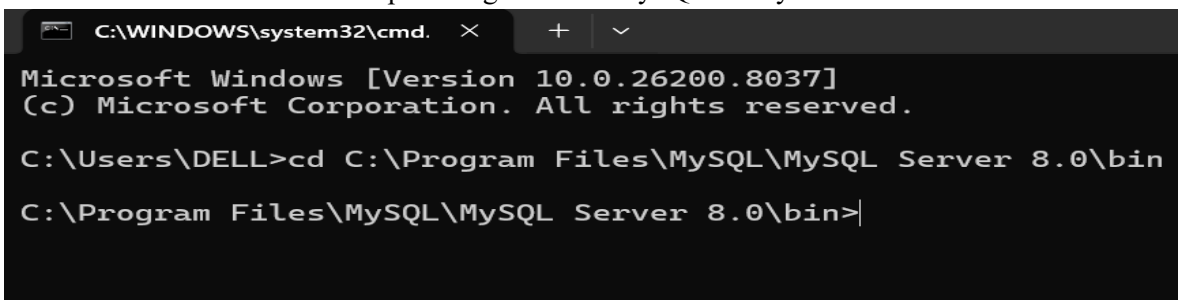


Task 12: After the configuration is completed, the Installation Complete message appears.

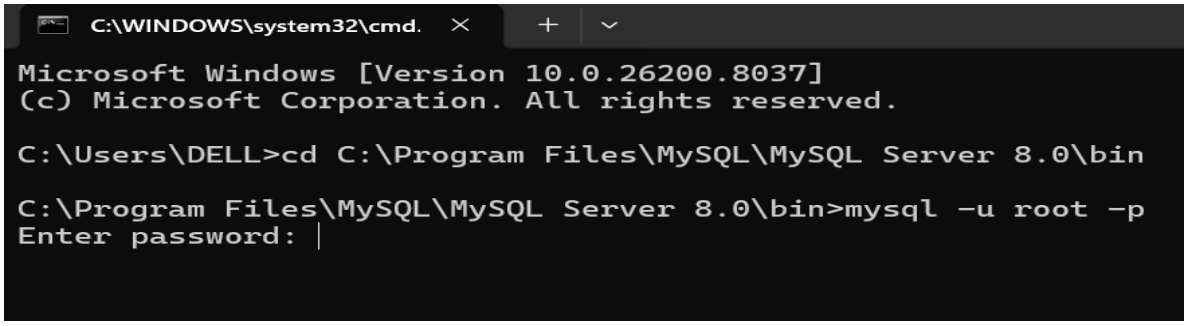


Task 13: Click Finish to complete the setup.

Task 14: In the Command Prompt. Navigate to the MySQL binary folder.



Task 15: Start the MySQL client.



Task 16: Enter the root password when prompted.

```
C:\WINDOWS\system32\cmd.
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: **************
```

Task 17: After entering the password, the MySQL Monitor starts and displays the welcome message for MySQL Community Server.

```
C:\WINDOWS\system32\cmd.
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: **************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.45 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> |
```

Task 18: To check the available databases, type the command: SHOW DATABASES.

```
C:\WINDOWS\system32\cmd.
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>cd C:\Program Files\MySQL\MySQL Server 8.0\bin

C:\Program Files\MySQL\MySQL Server 8.0\bin>mysql -u root -p
Enter password: **************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 12
Server version: 8.0.45 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current inp

mysql> SHOW DATABASE;
ERROR 1064 (42000): You have an error in your SQL syntax; check t
t syntax to use near 'DATABASE' at line 1
mysql> SHOW DATABASES;|
```

Task 22: The system then displays the list of databases available in the MySQL server.

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
4 rows in set (0.02 sec)

mysql> |
```

Conclusion

In this practical exercise, I learned how to download, install, and configure the MySQL Database Management System on a Microsoft Windows computer. The installation process involved selecting the required components such as MySQL Server, MySQL Workbench, and MySQL Shell, and configuring the server settings.

This practical activity helped me understand the fundamental steps involved in setting up a database system. I also learned how to access the MySQL server through the Command Prompt and execute basic SQL commands to interact with the database.

During the exercise, I encountered some challenges related to SQL syntax. For instance, when I initially attempted to check the list of databases using the command show database;, an error occurred because the correct SQL syntax requires the command SHOW DATABASES;. This mistake helped me realize the importance of writing SQL commands accurately and following the correct syntax rules.

By correcting the error and successfully executing the command, I was able to view the available databases in the MySQL server. This experience improved my understanding of how SQL commands work and how users interact with a database management system.
Overall, this exercise provided valuable hands-on experience with database installation, configuration, and basic database interaction. It also helped strengthen my understanding of the role of database systems in software development and data management.

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